

A Game-Theoretic Approach to Solving the Roman Domination Problem

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Abstract—The Roman domination problem is an important combinatorial optimization problem that is derived from an old story of defending the Roman Empire and now regains new significance in cyber space security, considering backups in the face of a dynamic network security requirement. In this paper, firstly, we propose a Roman domination game (RDG) and prove that every Nash equilibrium (NE) of the game corresponds to a strong minimal Roman dominating function (S-RDF), as well as a Pareto-optimal solution. Secondly, we show that RDG is an exact potential game, which guarantees the existence of an NE. Thirdly, we design a game-based synchronous algorithm (GSA), which can be implemented distributively and converge to an NE in $O(n)$ rounds, where n is the number of vertices. In GSA, all players make decisions depending on local information. Furthermore, we enhance GSA to be enhanced GSA (EGSA), which converges to a better NE in $O(n^2)$ rounds. Finally, we present numerical simulations to demonstrate that EGSA can obtain a better approximate solution in promising computation time compared with state-of-the-art algorithms.

Index Terms—Distributed algorithm, game theory, multi-agent system, potential game, Roman dominating function.

I. INTRODUCTION

THE minimum Roman domination (MinRD) problem was originated from an interesting historical story [1] and formally defined in [2]. Now, MinRD problem also has contemporary applications, especially in the field of server placements [3] and wireless sensor networks [4] in the face of a dynamic security setting. For example, in a wireless sensor network, a set of sensors are deployed at some nodes to monitor the neighboring environment. When a vacant node faces a security problem that needs to be fixed by moving a neighbor-

ing sensor to it, the moving sensor should leave behind at least one backup sensor. In other words, a node without a sensor has to be adjacent with a node deployed of at least two sensors. From an economic point of view, it is desired that the total number of sensors is as small as possible, under the condition that the above requirement is satisfied. This consideration leads to a MinRD problem.

The Roman domination problem was mathematically introduced in [2]. Before that, the definition of a Roman dominating function (RDF) was given implicitly in [1] and [5]. MinRD was proved to be NP-hard on general graphs [6] by a polynomial reduction from the 3-Satisfiability problem. In [3], a $(2 + 2 \ln n)$ -approximation algorithm is proposed for MinRD on general graphs, and a polynomial-time approximation scheme (PTAS) is designed for MinRD on planar graphs. In [7], it is proved that MinRD is NP-hard even on unit disk graphs, with a 5-approximation algorithm and a PTAS developed, making use of the geometry of unit disk graphs. In [8], a generalized Roman domination problem called the connected strong k -Roman dominating set problem is formulated, which is proved to be NP-hard on unit ball graphs, and a $6(k+2)$ -approximation algorithm is designed for unit ball graphs. There are also other variants of Roman domination problems [9]–[11], and many studies of the MinRD for special graphs [12]–[16].

Most algorithms developed in the above articles are centralized, that is, there is a central controller that manages all the processes. Such algorithms are typically vulnerable to cyber attacks. Furthermore, it is difficult to meet the flexibility and diversity requirements of real applications in large-scale networks with centralized algorithms. Therefore, distributed algorithms are more desirable, especially in multi-agent systems.

In a multi-agent system, every agent can make their own decision using current local information. The autonomy of agents eliminates the dependence on a central controller and greatly improves the anti-attack ability of the system. However, individual interests may have conflicts with social welfare, and thus a distributed algorithm may result in an unsatisfactory solution. Game theory is an effective method to coordinate such a conflict. In game theory, it is assumed that all players are selfish, rational and intelligent, who are only interested in maximizing their own benefits [17]. To align individual interests with social welfare, a crucial task is to set up suitable utility functions for the players such that their selfish

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behaviors can autonomously evolve into a satisfactory and stable collective behavior, where stability means that no player is willing to change his current strategy unilaterally. Such a stable state is called a Nash equilibrium (NE).

With the rapid development of multi-agent systems and large-scale networks, game theory has been widely used in the study of combinatorial optimization problems, for example to deal with the dominating set problem and its variants. In [18], a multi-domination game is studied and it is proved that every NE of the game is a minimal multi-dominating set, which is also a Pareto-optimal solution. In this game, a distributed algorithm is designed, where all players make decisions in a given order. Then, in [19], an independent domination game is formulated and it is proved that every NE is a minimal independent dominating set. Similarly, all players are required to make decisions only in a given order. Further, in [20] a connected domination game is investigated and it is proved that every NE is a minimal connected dominating set but not a Pareto-optimal solution. Later, in [21], a secure dominating game is studied and it is proved that every NE is a minimal secure dominating set which is also a Pareto-optimal solution. Furthermore, a distributed algorithm is designed, which allows all players to make decisions simultaneously and all players to use only local information. However, distributed algorithms for the Roman domination problem are rare and many studies on this problem are for special graphs. The present paper is perhaps the first one using game theory to study the Roman domination problem on general graphs. We propose a Roman domination game (RDG) and prove that every NE of the game is not only a minimal Roman dominating function (M-RDF) but also strong minimal (S-RDF), which possesses a local optimum property in a strong sensor. We design a game-based synchronous algorithm (GSA), which allows the players to make decisions simultaneously, all using local information. Furthermore, we enhance the GSA to the enhanced GSA (EGSA), which converges to a better NE than the GSA.

Another problem closely related to the subject of this paper is the minimum vertex cover (MinVC) problem. In [22], the MinVC is studied from the approach of the snowdrift game and a distributed algorithm is proposed. In [23], the minimum weight vertex cover (MinWVC) problem is investigated and an algorithm is designed based on an asymmetric game, which can find a vertex cover with smaller weight. In [24], [25], the MinWVC problem is solved using potential game theory and a distributed algorithm is proposed based on relaxed greedy algorithm with finite memory. Later, in [26] a population based game theoretic optimizer is designed, which combines learning with optimization. In [27], a weighted vertex cover game is proposed using a 2-hop adjustment scheme, so as to obtain a better solution. In addition to the above works, there are some reports using the game theory to study the coverage problems from different perspectives. For example, in [28], [29] cost sharing and strategy-proof mechanisms are adopted for set cover games. In [30], the core stability of the vertex cover game is studied. In [31], the core of a dominating set game is studied. In [32], the core of a connected dominating

set game is investigated. In [33], the price of anarchy and the computational complexity of a coverage game are analyzed.

In this paper, we focus on the MinRD problem in a multi-agent system. Main contributions of our paper are as follows:

1) We construct a game framework of multi-agent systems for the MinRD problem and prove the existence of NE for the RDG. Furthermore, we classify three types of RDF as S_{M-RDF} , S_{S-RDF} and S_{G-RDF} , show that $S_{G-RDF} \subseteq S_{ENE} \subseteq S_{NE} \subseteq S_{S-RDF} \subseteq S_{M-RDF} \subseteq S_{RDF}$, and prove that every NE is a Pareto-optimal solution, where S_{ENE} and S_{NE} are the set of enhanced NEs and the set of NEs, respectively.

2) We propose three algorithms for the RDG, named the game-based asynchronous algorithm (GAA), game-based synchronous algorithm (GSA) and enhanced game-based synchronous algorithm (EGSA). In GAA, an NE converges in $O(n)$ rounds of interactions and players make decisions depending on local information at most one hop away. In GSA, not only can it achieve the same performances as GAA, but also the computation can be realized distributedly, where all players make decisions simultaneously. In EGSA, a better solution than GSA can be obtained, at the expense of a longer running time $O(n^2)$.

3) Our numerical simulation shows that the proposed algorithms are better than the existing algorithms on random graphs. On random tree graphs, our algorithms yield results closer to the optimal solutions, whereas the gap between the result of our EGSA and the optimal solution is less than 0.01%.

The remaining parts of the paper are organized as follows. Section II introduces preliminaries in game theory and MinRD, and constructs an RDG. Section III provides a strict theoretical analysis for the game and explores the relationship between NE and the three types of RDF. Section IV describes the three algorithms in details and provides theoretical analysis on its convergence. Section V evaluates the performance of the three algorithms through extensive simulations. Section VI concludes the paper with some discussions on future work.

II. MODELLING ROMAN DOMINATION PROBLEM AS A GAME

A. Preliminaries

A game Γ is represented by $\Gamma = (V, \{S_i\}_{i=1}^n, \{u_i\}_{i=1}^n)$, where $V = \{v_1, v_2, \dots, v_n\}$ is the set of players, S_i is v_i 's strategy set, and u_i is v_i 's utility function. The strategy space of the game is $\Sigma = S_1 \times S_2 \times \dots \times S_n$. A strategy profile is an n -tuple $C = (c_1, c_2, \dots, c_n) \in \Sigma$. For player v_i , write $C = (c_i, C_{-i})$, where $C_{-i} = (c_1, \dots, c_{i-1}, c_{i+1}, \dots, c_n)$ indicates the strategies of those players except v_i . Let $u_i(C)$ denote the utility of v_i under strategy profile C . Players are assumed to be selfish, intelligent and rational, which means that the goal of every player is to maximize his own utility. The best response of player v_i to the current strategy profile C is

$$BR(v_i, C) = \arg \max \{u_i(c'_i, C_{-i}) : c'_i \in S_i\}.$$

A Nash equilibrium (NE) is a strategy profile C such that no player wants to deviate from C unilaterally, formally defined as follows.

Definition 1 (Nash equilibrium [34]): Given a game $\Gamma =$

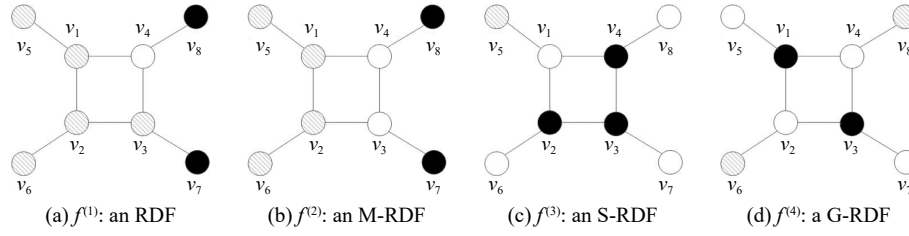


Fig. 1. An example to show the three types of RDFs.

$(V, \{S_i\}_{i=1}^n, \{u_i\}_{i=1}^n)$, a strategy profile $C^* = (c_1^*, c_2^*, \dots, c_n^*)$ is a Nash equilibrium (NE) if

$$u_i(c_i^*, C_{-i}^*) \geq u_i(c_i, C_{-i}^*) \text{ for any } v_i \in V \text{ and any } c_i \in S_i.$$

Notice that an NE is not necessarily a global optimal solution. Especially, for an NP-hard problem, in many cases, Pareto-optimal solutions are satisfactory.

Definition 2 (Pareto-optimal solution): Given a game $\Gamma = (V, \{S_i\}_{i=1}^n, \{u_i\}_{i=1}^n)$, a strategy profile $C' = (c'_1, \dots, c'_n)$ strictly dominates strategy profile $C = (c_1, \dots, c_n)$ if $u_i(C') \geq u_i(C)$ holds for all index $i \in \{1, \dots, n\}$ and there exists an index $j \in \{1, \dots, n\}$ with $u_j(C') > u_j(C)$. A strategy profile $C^* = (c_1^*, \dots, c_n^*)$ is a Pareto-optimal solution if there is no strategy profile which strictly dominates C^* .

Definition 3 (Roman dominating function (RDF)): Given a graph $G = (V, E)$, a function $f : V(G) \rightarrow \{0, 1, 2\}$ is an RDF of G if each vertex $v \in V$ with $f(v) = 0$ is adjacent to a vertex $u \in V$ with $f(u) = 2$.

RDF originates from an interesting historical story [1]: to maintain the safety of the Roman Empire, emperor Constantine adopted the strategy of “island-hopping” — moving troops from one island to a nearby island, but only when he could leave behind a large enough garrison to keep the first island secure. This strategy was also adopted by MacArthur during military operations in World War II. In the language of graph theory, RDF asks for a deployment of troops on some vertices such that any vertex without a troop must be adjacent with a vertex with at least two troops. Such a concern also occurs in the field of server placement in a wireless sensor network [3], [4] as shown at the beginning of this paper. It is desired that the number of servers is as small as possible, which leads to the MinRD problem defined as follows.

Definition 4 (Minimum Roman domination problem (MinRD)): Given a graph $G = (V, E)$, for any RDF f , let $\gamma_R(G, f) = \sum_{v \in V} f(v)$ be the weight of the RDF f . The goal of MinRD is to minimize $\gamma_R(G, f)$ over all RDFs f .

Denote by S_{RDF} the set of RDFs. In the following, we define three types of RDF.

Definition 5 (Global minimum Roman dominating function (G-RDF)): An RDF f^* of graph G is a G-RDF if $\gamma_R(G, f^*) = \min_{f \in S_{RDF}} \gamma_R(G, f)$. Let S_{G-RDF} be the set of G-RDFs.

Finding a G-RDF is the ultimate goal but too ambitious to be realized, because it is an NP-hard problem [6]. In fact, Padamutham and Palagiri [35] had shown that even for bipartite graphs, a G-RDF cannot be approximated within factor $(1 - \varepsilon) \ln |V|$ for any $\varepsilon > 0$ unless $NP \subseteq DTIME(|V|^{O(\log \log |V|)})$. So, a realistic hope is that the solution has a reasonably small size, say, instead of a minimum one, the solution should be minimal, that is, under the current deployment, no server can

be omitted to keep an RDF.

Definition 6 (Minimal Roman dominating function (M-RDF)): An RDF f of graph G is an M-RDF if there is no RDF f' of G satisfying $f'(v_i) = f(v_i) - 1$ for some $v_i \in V$ and $f'(v_j) = f(v_j)$ for any $j \neq i$. Let S_{M-RDF} be the set of M-RDFs.

The following concept of S-RDF further reduces the size of M-RDF from a local search point of view.

Definition 7 (Strong minimal Roman dominating function (S-RDF)): An M-RDF f is an S-RDF if there is no RDF f' with $\gamma_R(G, f') < \gamma_R(G, f)$ satisfying $f'(v_i) = 2$ for some $v_i \in V$, $f'(v_j) = 0$ for any $v_j \in N_i$ with $f(v_j) = 1$ and $f'(v_k) = f(v_k)$ for any other v_k . Let S_{S-RDF} be the set of S-RDFs.

Fig. 1 shows three different types of RDFs. For simplicity of statement, call a vertex v_i to be white, gray, or black if $f(v_i) = 0, 1$ and 2 , respectively. In Fig. 1, $f^{(1)}, f^{(2)}, f^{(3)}$ and $f^{(4)}$ are all RDFs, because every white vertex is adjacent to at least one black vertex. But $f^{(1)}$ is not an M-RDF, because f' defined in the following way is also an RDF: $f'(v_3) = f^{(1)}(v_3) - 1$ and $f'(v_j) = f^{(1)}(v_j)$ for any $v_j \neq v_3$. It can be demonstrated that $f^{(2)}, f^{(3)}$ and $f^{(4)}$ are all M-RDFs. For example, in Fig. 1(b), after changing a gray vertex v_i with $i \in \{1, 2, 5, 6\}$ to a white vertex, there is no black vertex adjacent to v_i ; after changing a black vertex v_7 (or v_8) to a gray vertex, there is no black vertex adjacent to v_3 (or v_4). In both cases, f' is no longer an RDF. Note that $f^{(2)}$ is not an S-RDF, because f' defined in the following way is an RDF with $\gamma_R(G, f') = 7 < 8 = \gamma_R(G, f^{(2)})$: $f'(v_1) = 2$, $f'(v_j) = 0$ for $j = 2, 5$ and $f'(v_k) = f(v_k)$ for $k = 3, 4, 6, 7, 8$. It can be demonstrated that both $f^{(3)}$ and $f^{(4)}$ are S-RDFs. Furthermore, $f^{(4)}$ is a G-RDF, while $f^{(3)}$ is not.

To facilitate reading, main notations are summarized in Table I.

B. Roman Domination Game

In this subsection, the RDG is introduced. Given a graph $G = (V, E)$, each vertex $v \in V$ can be viewed as a player. For a player v_i , his strategy $c_i = 0, 1$ or 2 indicates that $f(v_i) = 0, 1$ or 2 , respectively.

It should be noted that RDF is *monotonic* in the sense that if C is a profile corresponding to an RDF, C' is another profile with $C' \geq C$ (that is $c'_i \geq c_i$ for every i), thus C' also corresponds to an RDF; and if C is not an RDF, $C' \leq C$ is not an RDF either.

Assuming that all players are selfish, intelligent and rational, they will not consider the benefits of the other players while seeking to maximize their own benefits. The critical task of the game is to design a good utility function for the players, such that a stable and fairly good social state can be

TABLE I
LIST OF NOTATIONS

Notation	Meaning
Γ	The game
n	The number of vertices, also the number of players
v_i	Both a vertex in the graph and a player in the game
V	The set of players or vertices $V = \{v_1, v_2, \dots, v_n\}$
S_i	$S_i = \{0, 1, 2\}$ is the strategy set of player v_i
Σ	$\Sigma = S_1 \times \dots \times S_n$ is the strategy space
C	$C = (c_1, c_2, \dots, c_n) \in \Sigma$ is a strategy profile
$u_i(C)$	The utility of v_i under strategy profile C
$d(i, j)$	The length of a shortest path between vertex v_i and v_j
N_i	$N_i = \{v_j \in V : v_j \text{ is adjacent to } v_i\}$ is the neighbor set of v_i
$\bar{N}_i$	$\bar{N}_i = N_i \cup \{v_i\}$ is the closed neighbor set of v_i
$N_{i,k}$	$N_{i,k} = \{v_j \in V : 0 < d(i, j) \leq k\}$ is the k -hop neighbor set of v_i
$\bar{N}_{i,k}$	$\bar{N}_{i,k} = N_{i,k} \cup \{v_i\}$ is the closed k -hop neighbor set of v_i
$N_i^j(C)$	$N_i^j(C) = \{v_k \in N_i : c_k = j\}$

reached through cooperation and competition among players, where the goodness of the state is measured by the following criteria.

1) *Self-Stability*: Starting from any initial state, the game can end up in a Nash equilibrium which corresponds to an RDF.

2) *Solution Quality*: The weight of the RDF corresponding to the NE should be reasonably small. Since the computation of a G-RDF is NP-hard even using centralized algorithms, one cannot hope for a minimum solution of MinRD in reasonable time. An alternative basic requirement is that the computed RDF should be minimal, that is, decreasing one bit of the f -value of any vertex v_i with $f(v_i) \neq 0$ will no longer be an RDF. Furthermore, it is desired to maximize social welfare, for which Pareto optimality is an important indicator.

3) *Efficient Execution*: The time for the game to reach an NE should be polynomial in the size of the input. The information used for players should be local. Moreover, a distributed algorithm is preferred.

For a strategy profile $C = (c_1, c_2, \dots, c_n)$, we define the utility function of v_i as

$$u_i(C) = g_i(C) + q_i(C) \quad (1)$$

with

$$g_i(C) = -\lambda_1 c_i^2 \quad (2)$$

$$q_i(C) = -\lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C) \quad (3)$$

where

$$m_j(C) = \begin{cases} 1, & \bar{N}_j^2(C) = \emptyset \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

and λ_1, λ_2 are constants satisfying $0 < \frac{2}{3}\lambda_2 < \lambda_1 < \frac{3}{4}\lambda_2$.

In the following, for a strategy profile C , we designate vertex v_i as strongly dominated if there is a vertex $v_j \in \bar{N}_i$ with $c_j = 2$, otherwise we designate v_i as free. Note that a player

$$v_i \text{ is strongly dominated if and only if } m_i(C) = 0. \quad (5)$$

Thus, we can see that a black vertex is strongly dominated by itself. A vertex is dominated if it is either black, or gray, or, it could be a strongly dominated white vertex. Note that C is an RDF if and only if all vertices are dominated.

Furthermore, assume that G has no isolated vertex. In fact, if v_i is an isolated vertex, in order for C to be an M-RDF, v_i must be gray. From the game point of view, the best response of v_i is also gray, since $u_i(c_i = 0, C_{-i}) = -2\lambda_2$, $u_i(c_i = 1, C_{-i}) = -\lambda_1 - \lambda_2$ and $u_i(c_i = 2, C_{-i}) = -4\lambda_1$. Thus, such trivial case will be ignored in the later discussion.

To obtain some information about u_i , consider the example in Fig. 2. The term $-g_i$ can be viewed as the cost of v_i that only depends on v_i 's own strategy. A black vertex v_i should pay $4\lambda_1$ because $g_i(C) = -4\lambda_1$, a gray vertex v_j should pay λ_1 because $g_j(C) = -\lambda_1$ and a white vertex v_k pays nothing because $g_k(C) = 0$. The term q_i reflects the interactive influence between v_i and its neighbors.

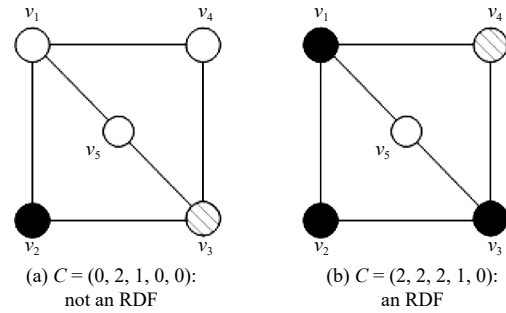


Fig. 2. An example illustrating the ideas behind the definition of utility function u_i .

For a strategy profile C which is not an RDF, the design of the utility function encourages C to evolve for an RDF. In Fig 2(a), $C = (0, 2, 1, 0, 0)$ is not an RDF. Every free white vertex v_i is inclined to become gray, even though it is subject to a cost of λ_1 from g_i , the interactive gain from its neighbors is high enough to cover its cost. For example, v_4 is a free white vertex, $u_4(C) = -2\lambda_2$ and $u_4(c'_4 = 1, C_{-4}) = -\lambda_1 - \lambda_2$. Thus, $u_4(c'_4 = 1, C_{-4}) - u_4(C) = \lambda_2 - \lambda_1 > 0$. The utility function also encourages a dominated vertex v_i to be black if such a change can help some free neighbors become strongly dominated. For example, v_1 is a dominated vertex and both v_4 and v_5 are v_1 's free neighbors. Because $u_1(C) = -4\lambda_1$ and $u_1(c_1 = 2, C_{-1}) = -4\lambda_1$, we have $u_1(c_1 = 2, C_{-1}) - u_1(C) = 4(\lambda_1 - \lambda_2) > 0$. So, v_1 has an incentive to change his strategy from 0 to 2, and at the same time, both v_4 and v_5 become strongly dominated vertices.

For a strategy profile C which is already an RDF, the design of the utility function aims to reduce the weight of the RDF. For example, in Fig 2(b), $C = (2, 2, 2, 1, 0)$ is an RDF. The utility function encourages black vertices to become white if such a change does not cause other dominated vertices to become free. For example, v_2 is a black vertex, if v_2 changes its strategy from $c_2 = 2$ to $c'_2 = 0$, then v_2 is also a dominated vertex and does not cause other dominated vertex to become free. In view of utility, $u_2(C) = -4\lambda_1 < u_2(c'_2 = 0, C_{-2}) = 0$. A similar incentive exists for a gray vertex. For example, v_4 is a gray vertex; if v_4 changes its strategy from $c_4 = 1$ to $c'_4 = 0$, then v_4

is still a dominated vertex and does not cause other dominated vertices to become free. In terms of the utility function, because $u_4(C) = -\lambda_1 < u_4(c'_4 = 0, C_{-4}) = 0$, v_4 is willing to become white.

III. THEORETICAL ANALYSIS

In this section, the theoretical properties of the RDG designed in the above section will be analyzed. In the following, we shall use C to denote both a strategy profile $(c_1, \dots, c_n) \in \Sigma$ and the corresponding function f with $f(v_i) = c_i$, ($i = 1, \dots, n$). It is assumed that a player is willing to change his strategy only when he can be *strictly* better off, that is, when his utility becomes strictly larger after changing his current strategy.

A. Nash Equilibrium

Nash equilibrium [34] is a stable state in the game. In this subsection, the properties of NE in RDG will be analyzed. For ease of statements, some preliminary results are given as observations and lemmas.

Observation 1: By the definition of $m_j(C)$, a) changing the strategy of v_i between 0 and 1 does not affect $m_j(C)$ for any v_j ; b) changing the gray vertex to white does not result in other vertices being strongly dominated or free.

Lemma 1: For any player v_i with $c_i = 2$ under strategy profile C , $u_i(C) = -4\lambda_1$.

Proof: Every player $v_j \in \bar{N}_i$ is strongly dominated because $c_i = 2$, and thus $m_j(C) = 0$ by observation (5). The lemma follows from the definition of the utility function. ■

Lemma 1 states that players will directly receive a utility value $-4\lambda_1$ after choosing strategy 2, which is independent of the strategies of the other players.

Lemma 2: In any strategy profile C , for any player $v_i \in V$, $u_i(c_i = 1, C_{-i}) > u_i(c'_i = 0, C_{-i})$ if and only if $m_i(C) = 1$.

Proof: Let $C = (c_i = 1, C_{-i})$ and $C' = (c'_i = 0, C_{-i})$. Then

$$u_i(C) = -\lambda_1 - \lambda_2 m_i(C) - \lambda_2 \sum_{v_j \in N_i} (2 - c_j) m_j(C)$$

$$u_i(C') = -2\lambda_2 m_i(C') - \lambda_2 \sum_{v_j \in N_i} (2 - c_j) m_j(C').$$

By Observation 1-a), $m_j(C) = m_j(C')$, $\forall v_j \in V$. Thus $u_i(C) - u_i(C') = \lambda_2 m_i(C) - \lambda_1 > 0$ if and only if $m_i(C) = 1$ by $\lambda_2 > \lambda_1$. ■

Lemma 2 provides a necessary and sufficient condition for a player to judge whether strategy 1 is better than strategy 2.

Theorem 1: Every NE of the RDG is an RDF.

Proof: Suppose this is not true. Let C be an NE which is not an RDF. Then, $\exists v_i \in V$ with $c_i = 0$ and $N_i^2(C) = \emptyset$. It follows that $m_i(C) = 1$ by the definition of $m_i(C)$. Let C' be $(c'_i = 1, C_{-i})$. By Lemma 2, $u_i(c'_i = 1, C_{-i}) > u_i(c_i = 0, C_{-i})$. It contradicts the fact that C is an NE. Hence, C is an RDF. ■

Theorem 2: Every NE of an RDG is an M-RDF.

Proof: Suppose C is an NE, but is not an M-RDF. By Theorem 1, it suffices to consider the minimality of C . Suppose $v_i \in V$ is a vertex with $c_i \geq 1$ such that $C' = (c'_i = c_i - 1, C_{-i})$ is still an RDF. There are two cases to be considered.

Case 1: $c_i = 1$

In this case, the assumption of $C' = (c'_i = 0, C_{-i})$ is still an RDF which implies that $N_i^2(C') = N_i^2(C) \neq \emptyset$, and thus $m_i(C') = m_i(C) = 0$. By Lemma 2, $u_i(c'_i = 0, C_{-i}) > u_i(c_i = 1, C_{-i})$, which is contradictory to the assumption that C is an NE.

Case 2: $c_i = 2$

In this case, $C' = (c'_i = 1, C_{-i})$ is also an RDF. If $N_i^1(C) \neq \emptyset$, let $v_j \in N_i^1(C)$. Then $(c'_j = 0, C_{-j})$ is still an RDF by Observation 1 (b). This situation is reduced to Case 1 (with v_j playing the role of v_i). Hence, suppose $N_i^1(C) = \emptyset$. The assumption that $C' = (c'_i = 1, C_{-i})$ is an RDF implies that for any $v_j \in N_i^0(C')$, there exists a $v_k \in N_j^2(C')$, and thus $m_j(C') = 0$ by observation (5). Hence, $u_i(c'_i = 1, C_{-i}) = -\lambda_1 - \lambda_2 m_i(C') \geq -\lambda_1 - \lambda_2$ (note that $m_i(C') \leq 1$). Because $u_i(c_i = 2, C_{-i}) = -4\lambda_1$ by Lemma 1, $u_i(c'_i = 1, C_{-i}) - u_i(c_i = 2, C_{-i}) \geq 3\lambda_1 - \lambda_2 > \lambda_2 > 0$ by the assumption of $0 < \frac{2}{3}\lambda_2 < \lambda_1$, which is a contradiction to that C is an NE. ■

Corollary 1: In any NE C , the following three conditions are satisfied:

- i) No gray vertices are adjacent to black vertices,
- ii) Any white vertex v_i is adjacent to at least one black vertex,
- iii) If a vertex v_i is gray, then $m_i(C) = 1$.

Proof:

Property i) follows from the minimality of C and Observation 1-b). In fact, if there exists a gray vertex v_i adjacent to a black vertex v_j , then $C' = (c'_i = 0, C_{-i})$ is also an RDF, contradicting the minimality of C .

Property ii) is a direct result of the definition of RDF and Theorem 1.

Property iii) follows from property i), because for any gray vertex v_i , $N_i^2(C) = \emptyset$, and thus $m_i(C) = 1$ by the definition of $m_j(C)$. ■

Theorem 3: Every NE of an RDG is an S-RDF.

Proof: Suppose C is an NE. By Theorem 2, C is an M-RDF. Thus, if C is not an S-RDF, then $\exists$ an RDF C' with $\gamma_R(G, C') < \gamma_R(G, C)$ satisfying $c'_i = 2$ for some $v_i \in V$, $c'_j = 0$ for any $v_j \in N_i$ with $c_j = 1$, and $c'_k = c_k$ for the other $v_k \notin N_i$.

If $c_i = 2$, then by Corollary 1-i), no vertex in N_i has value 1. So, $C' = C$, a contradiction to $\gamma_R(G, C') < \gamma_R(G, C)$.

If $c_i = 1$, then by $\gamma_R(G, C') < \gamma_R(G, C)$, there are at least two vertices $v_j, v_k \in N_i$ with $c_j = c_k = 1$ and $c'_j = c'_k = 0$. By Corollary 1-iii), $m_i(C) = m_j(C) = m_k(C) = 1$. Then $u_i(C) \leq -\lambda_1 - 3\lambda_2$. Consider a strategy profile $C'' = (c''_i = 2, C_{-i})$. By Lemma 1, $u_i(C'') = -4\lambda_1$. It follows that $u_i(C'') - u_i(C) \geq 3\lambda_2 - 3\lambda_1 > 0$, contradicting that C is an NE.

If $c_i = 0$, then $\gamma_R(G, C') < \gamma_R(G, C)$ implies that, there are at least three vertices $v_j, v_k, v_l \in N_i$ with $c_j = c_k = c_l = 1$ and $c'_j = c'_k = c'_l = 0$. Similarly as above, $m_j(C) = m_k(C) = m_l(C) = 1$, and thus $u_i(C) \leq -3\lambda_2$, while $C'' = (c''_i = 2, C_{-i})$ has $u_i(C'') = -4\lambda_1$. Hence $u_i(C'') - u_i(C) \geq 3\lambda_2 - 4\lambda_1 > 0$, contradicting that C is an NE. ■

In the above proof, one can see that the two substructures in Fig. 3 do not appear in an NE.

Corollary 2: No NE of an RDG has the two “bad” substructures shown in Fig. 3.

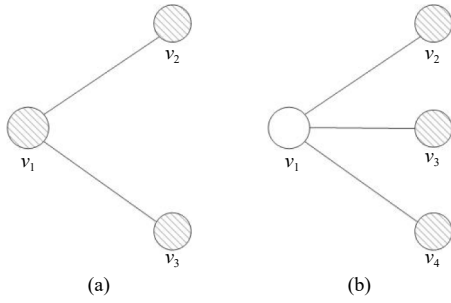


Fig. 3. Some “bad” substructures ((a) A gray vertex having at least two gray neighbors; (b) A white vertex having at least three gray neighbors).

In fact, the two substructures do not appear in any S-RDF.

Lemma 3: For any S-RDF C , the two substructures shown in Fig. 3 cannot appear.

Proof: Define

$$C' = \begin{cases} 2, & v_1 \\ 0, & v_j \in N_1 \wedge (c_j \neq 2) \\ c_j, & \text{other } v_j. \end{cases}$$

Then, for both cases in Fig. 3, it can be demonstrated that C' is also an RDF and $\gamma_R(G, C') \leq \gamma_R(G, C) - 1 < \gamma_R(G, C)$, and the lemma follows from the definition of S-RDF. ■

By Theorem 3, $S_{NE} \subseteq S_{S-RDF}$, where S_{NE} is the set of NEs of the RDG. Next, it can be proven that $S_{G-RDF} \subseteq S_{NE}$. This is a result of Lemma 11 (when we study the algorithmic aspect of the game). The following is an existence proof. It is included here for theoretical completeness. For this purpose, first a lemma is established.

Lemma 4: In any S-RDF, there is no player $v_i \in V$ who is willing to increase his c_i -value unilaterally.

Proof: Let C be an S-RDF. If $c_i = 2$, then v_i cannot increase its value.

Suppose v_i increases its value from $c_i = 1$ to $c'_i = 2$. Because C is an S-RDF, every white vertex is adjacent to at least one black vertex. So, $m_j(C) = 0$ for any v_j with $c_j = 0$. Hence, $u_i(C) = -\lambda_1 - \lambda_2 \sum_{v_j \in N_i^1(C)} (2 - c_j) m_j(C)$. By Lemma 3, there is at most one gray vertex $v_k \in N_i^1(C)$. Hence, $u_i(C) \geq -\lambda_1 - 2\lambda_2$. Since $u_i(c'_i = 2, C_{-i}) = -4\lambda_1$ by Lemma 1, $u_i(C) - u_i(c'_i = 2, C_{-i}) \geq 3\lambda_1 - 2\lambda_2 > 0$.

Suppose v_i increases its value from $c_i = 0$ to $c'_i = 1$. Because C is an S-RDF, $N_i^2(C) \neq \emptyset$, and thus $m_i(C) = 0$. By Lemma 2, $u_i(C) > u_i(c'_i = 1, C_{-i})$.

Suppose v_i increases its value from $c_i = 0$ to $c'_i = 2$. Because C is an S-RDF, $m_j(C) = 0$ for any v_j with $c_j = 0$. By Lemma 3, there are at most two gray vertices in $N_i^1(C)$. So, $u_i(C) \geq -2\lambda_2$. Since $u_i(c'_i = 2, C_{-i}) = -4\lambda_1$ by Lemma 1, $u_i(C) - u_i(c'_i = 2, C_{-i}) \geq 4\lambda_1 - 2\lambda_2 > 0$.

In any case, v_i is not willing to increase his strategy from c_i to c'_i . ■

Note that Lemma 4 implies that in an S-RDF, no player is willing to “increase” value, but “decreasing” value is still possible.

The next lemma gives some properties of G-RDF, which will be used to show that any G-RDF is an NE. For a strategy profile C , a vertex v_j is uniquely strongly dominated by v_i if v_i

is the unique vertex in $\bar{N}_j$ with c -value 2. Note that this definition allows v_i to be v_j itself.

Lemma 5: Let C be a G-RDF, the following two properties hold:

- i) No gray vertex is adjacent to a black vertex.
- ii) For any black vertex v_i , there are at least two vertices in $\bar{N}_i$ which are uniquely strongly dominated by v_i .

Proof: Suppose C has a gray vertex v_i which is adjacent to a black vertex. Then, v_i is still a dominated vertex if its status is changed from gray to white. By Observation 1-b), the status of the other vertices (dominated or not) are not affected by such a change. Since all vertices are dominated in C , so are all vertices in $C' = (c'_i = 0, C_{-i})$. But, then, C' is an RDF with smaller weight than C , contradicting the fact that C is a G-RDF. Property i) is proven.

Consider a black vertex v_i . By Property i), all neighbors of v_i are either black or white. Since it has been assumed that G has no isolation vertices, one has $N_i \neq \emptyset$. If all neighbors of v_i are black, then $C' = (c'_i = 0, C_{-i})$ is an RDF with $\gamma_R(G, C') = \gamma_R(G, C) - 2$, contradicting the fact that C is a G-RDF. So, $|N_i^0(C)| \geq 1$. If every vertex in $N_i^0(C)$ has another black neighbor, then $C' = (c'_i = 1, C_{-i})$ is an RDF with $\gamma_R(G, C') = \gamma_R(G, C) - 1$, which is also a contradiction. So, there is at least one vertex in $N_i^0(C)$, say v_j , which is uniquely strongly dominated by v_i . If all neighbors of v_i are white, then v_i is a vertex which is strongly dominated by itself, and Property ii) is satisfied. Next, suppose v_i has a black neighbor. If v_j is the only vertex in $N_i^0(C)$ which is uniquely strongly dominated by v_i , then $C' = (c'_i = 0, c'_j = 1, C_{-[i,j]})$ is an RDF with $\gamma_R(G, C') = \gamma_R(G, C) - 1$, again a contradiction. So, in this case, there are at least two vertices in $N_i^0(C)$ which are uniquely strongly dominated by v_i . Property ii) is proved. ■

Now, we are ready to prove $S_{G-RDF} \subseteq S_{NE}$.

Theorem 4: In an RDG, every G-RDF is an NE.

Proof: Suppose C is a G-RDF. Because a G-RDF is also an S-RDF, by Lemma 4, no player is willing to increase his c -value. So, to show that C is an NE, it suffices to show that no player v_i is willing to decrease his c -value from c_i to $c'_i < c_i$.

If $c_i = 2$, then $u_i(C) = -4\lambda_1$, and by Lemma 5-ii), there are at least two vertices $v_j, v_k \in \bar{N}_i$ which are uniquely strongly dominated by v_i . Reducing $c_i = 2$ to $c'_i = 1$ or $c'_i = 0$ results in a strategy profile C' . Note that v_j and v_k are no longer strongly dominated in C' , and thus $m_j(C') = m_k(C') = 1$. If $v_i \in \{v_j, v_k\}$, then the other vertex in $\{v_j, v_k\}$ is white by Lemma 5-i), and thus $u_i(c'_i = 1, C_{-i}) \leq -\lambda_1 - 3\lambda_2$, $u_i(c'_i = 0, C_{-i}) \leq -4\lambda_2$. If $v_i \notin \{v_j, v_k\}$, then both v_j, v_k are white, and thus $u_i(C') \leq -4\lambda_2$. In any case, $u_i(C) > u_i(C')$, and v_i is not willing to change.

If $c_i = 1$, then by Lemma 5 (i), $\bar{N}_i^2(C) = \emptyset$. It follows that $m_i(C) = 1$. So, $u_i(C) > u_i(c'_i = 0, C_{-i})$ by Lemma 2, and thus v_i is not willing to change. ■

By Definitions 3, 6 and 7, $S_{S-RDF} \subseteq S_{M-RDF} \subseteq S_{RDF}$. Combining this with Theorems 3 and 4, one has the following corollary.

Corollary 3: In an RDG, $S_{G-RDF} \subseteq S_{NE} \subseteq S_{S-RDF} \subseteq S_{M-RDF} \subseteq S_{RDF}$.

The relationship among these sets is illustrated in Fig. 4.

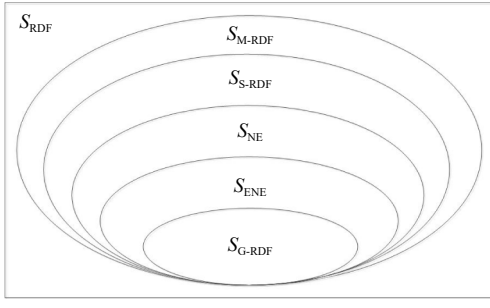


Fig. 4. Relationship among different types of RDF, where S_{ENE} is the set of ENEs obtained by Algorithm 3 defined in Section IV.

From Fig. 1, it can be seen that S_{S-RDF} might be a proper subclass of S_{M-RDF} . Next, it will be shown that the difference can be very large. For this purpose, define a parameter ρ_n as follows, as a measure of how big the difference is.

Definition 8: Let $\mathbb{G}$ be the set of all graphs on n vertices. Define

$$\rho_n = \max_{G \in \mathbb{G}} \frac{\min_{C' \in S_{M-RDF} \setminus S_{S-RDF}} \gamma_R(G, C')}{\max_{C'' \in S_{S-RDF}} \gamma_R(G, C''')}.$$

Consider the example in Fig. 5. It can be checked that $S_{M-RDF} \setminus S_{S-RDF} = \{C_1, C_2\}$, and $S_{S-RDF} = \{C_3\}$. Hence, $\rho_n \geq n/2$.

It has been proved that $S_{NE} \subseteq S_{S-RDF}$. The example in Fig. 6 shows that the inclusion might be proper. In fact, the strategy profile C in Fig. 6(a) satisfies $C \in S_{S-RDF} \setminus S_{NE}$, because $u_2(C) = -4\lambda_1 < -2\lambda_2 = u_2(C'_2 = 0, C_{-2})$. Note that C' is both an NE and an S-RDF.

B. Potential Game

In this subsection, it will be shown that the proposed RDG is an exact potential game, and thus NEs exist. Furthermore, an NE can be reached in linear rounds of interactions if the players are allowed to determine their strategies one by one.

Definition 9 (Exact potential game [36]): Call $\Gamma = (V; \{S_i\}_{i=1}^n; \{u_i\}_{i=1}^n)$ an exact potential game if there exists a potential function $\pi: \Sigma \rightarrow \mathbb{R}$ such that for any player $v_i \in V$ and any $c_i, c'_i \in S_i$, $C_{-i} \in S_{-i}$, the following equality holds:

$$\pi(c_i, C_{-i}) - \pi(c'_i, C_{-i}) = u_i(c_i, C_{-i}) - u_i(c'_i, C_{-i}).$$

Lemma 6: The proposed RDG is an exact potential game.

Proof: It will be proved that the following function is a potential function:

$$\pi(C) = -\lambda_1 \sum_{j=1}^n c_j^2 - \lambda_2 \sum_{j=1}^n (2 - c_j) m_j(C).$$

Denote the two terms of $\pi(C)$ as $\pi^{(1)}(C)$ and $\pi^{(2)}(C)$, respectively.

Let $C = (c_i, C_{-i})$ and $C' = (c'_i, C_{-i})$ be two strategy profiles before and after some v_i changes its strategy from c_i to c'_i . It can be verified that

$$\pi^{(1)}(C) - \pi^{(1)}(C') = \lambda_1(c_i'^2 - c_i^2) = g_i(C) - g_i(C'). \quad (6)$$

Note that changing the status of v_i does not affect the m -values of those vertices outside of $\bar{N}_i$, so one has

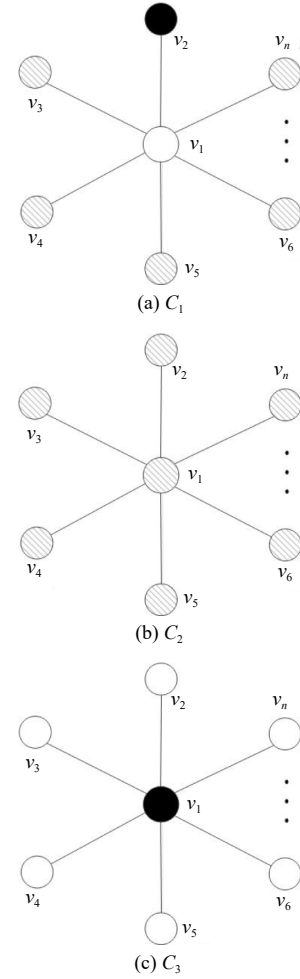


Fig. 5. An example showing that the difference between $S_{M-RDF} \setminus S_{S-RDF}$ and S_{S-RDF} can be very large.

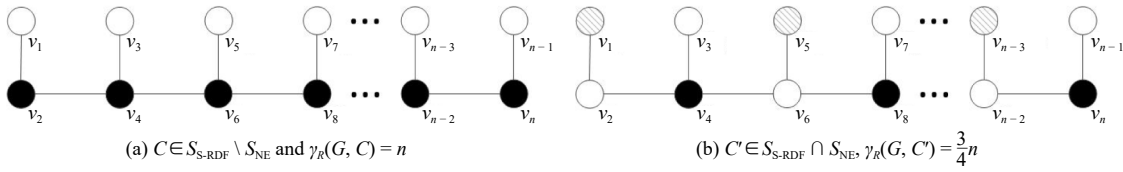
$$\begin{aligned} \pi^{(2)}(C) - \pi^{(2)}(C') &= -\lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C) \\ &\quad + \lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c'_j) m_j(C') \\ &= q_i(C) - q_i(C'). \end{aligned} \quad (7)$$

Combining (6) and (7), gives $\pi(C) - \pi(C') = u_i(C) - u_i(C')$. ■

As a consequence of Lemma 6, NEs exist. Furthermore, as shown by the following theorem, an NE can be reached in linear time steps of asynchronous interactions, where “asynchronous” means that players determine their strategies one by one, and thus in every round, although there are many players who are willing to change, only one actually takes action.

Theorem 5: Starting from any initial state, the number of iterations needed for an RDG to reach an NE is at most $\frac{(4\lambda_1 + 2\lambda_2)n}{\min\{3\lambda_1 - 2\lambda_2, -4\lambda_1 + 3\lambda_2\}} = O(n)$.

Proof: As long as C is not an NE, there is a player v_i who is willing to change his strategy from c_i to c'_i . Denote $\Delta u_i = u_i(C') - u_i(C)$, where $C' = (c'_i, C_{-i})$. Because v_i is willing to change only when he can be strictly better off. So, $\Delta u_i > 0$. Then, by Lemma 6, $\pi(C') - \pi(C) = u_i(C') - u_i(C) > 0$, that is, in every round of interaction, the potential function π is

Fig. 6. An example showing that $S_{S-RDF} \setminus S_{NE} \neq \emptyset$.

strictly increasing. To prove the theorem, it is needed to estimate the range of π and positive lower bounds of Δu_i (that is, the lower bounds of Δu_i under the condition that $\Delta u_i > 0$).

Clearly, $-(4\lambda_1 + 2\lambda_2)n \leq \pi(C) \leq 0$. The positive lower bounds for Δu_i , in different cases of c_i and c'_i , are shown in Table II. For example, when $c_i = 2$ and $c'_i = 0$, $u_i(C) = -4\lambda_1$ by Lemma 1 and $u_i(C'_i = 0, C_{-i}) = -\lambda_2 \sum_{v_j \in \tilde{N}_i} (2 - c_j)m_j(C')$. Because $4\lambda_1 - 2\lambda_2 > 0$ and $4\lambda_1 - 3\lambda_2 < 0$, the positive value of $u_i(C'_i = 0, C_{-i}) - u_i(C)$ in this case must be $\geq 4\lambda_1 - 2\lambda_2$. Using this argument for all cases, the estimations in Table II are obtained (see Appendix A for a detailed proof). Then, it can be verified that $\min\{3\lambda_1 - 2\lambda_2, -4\lambda_1 + 3\lambda_2\}$ is a positive lower bound of Δu_i . Hence, the number of iterations is at most $\frac{(4\lambda_1 + 2\lambda_2)n}{\min\{3\lambda_1 - 2\lambda_2, -4\lambda_1 + 3\lambda_2\}}$, which is $O(n)$ since λ_1 and λ_2 are constants satisfying $\frac{2}{3}\lambda_2 < \lambda_1 < \frac{3}{4}\lambda_2$. ■

TABLE II
ESTIMATION OF THE POSITIVE LOWER BOUNDS OF Δu_i

c_i	c'_i	$\min\{\Delta u_i, \Delta u_i > 0\}$
2	0	$4\lambda_1 - 2\lambda_2$
2	1	$3\lambda_1 - 2\lambda_2$
1	0	λ_1
0	1	$-\lambda_1 + \lambda_2$
1	2	$-3\lambda_1 + 3\lambda_2$
0	2	$-4\lambda_1 + 3\lambda_2$

C. Pareto Optimality

In the above subsection, the existence of NE and the quality of RDF corresponding to NE are discussed. In this subsection, the NE will be further analyzed from the perspective of social welfare, and it will be shown that NE has Pareto optimality.

Theorem 6: Every NE of an RDG is a Pareto-optimal solution.

Proof: Suppose $C = (c_1, \dots, c_n)$ is an NE but is not a Pareto-optimal solution. Then, there is a strategy profile $C' = (c'_1, \dots, c'_n) \neq C$ such that

$$u_i(C') \geq u_i(C) \text{ for any } i \in \{1, \dots, n\},$$

$$\exists j \in \{1, \dots, n\} \text{ with } u_j(C') > u_j(C). \quad (8)$$

Because C is an NE, by Corollary 1 and Corollary 2, for any $i \in \{1, \dots, n\}$, one has

$$u_i(c_i = 1, C_{-i}) = -\lambda_1 - \lambda_2 \text{ or } -\lambda_1 - 2\lambda_2 \quad (9)$$

$$u_i(c_i = 0, C_{-i}) = 0, -\lambda_2, \text{ or } -2\lambda_2. \quad (10)$$

Claim 1: For any v_i with $c_i \neq 2$, one has $c'_i \neq 2$.

If $c'_i = 2$, then $u_i(C') = -4\lambda_1$ by Lemma 1. By (9) and (10), $u_i(C) \geq -\lambda_1 - 2\lambda_2$. So, $u_i(C) > u_i(C')$, contradicting condition (8).

Claim 2: For any v_i with $c_i = 1$, one has $c'_i = 1$.

By Claim 1, one has $c'_i = 1$ or 0. If the claim is not true, then $c'_i = 0$. Consider the two possible values for $u_i(C)$ as shown in (9).

If $u_i(C) = -\lambda_1 - \lambda_2$, then by Corollary 1, in strategy profile C , v_i is the unique gray vertex in $\tilde{N}_i$ and all the vertices in N_i are white. By condition (8), $u_i(C') \geq -\lambda_1 - \lambda_2$. Thus, one must have $N_i^2(C') \neq \emptyset$, otherwise $u_i(C') \leq -2\lambda_2 m_i(C') = -2\lambda_2 < -\lambda_1 - \lambda_2$. Let v_k be a vertex in $N_i^2(C')$. Then, $u_k(C') = -4\lambda_1$ by Lemma 1. Since v_k is white in C , by (10), one has $u_k(C) \geq -2\lambda_2$. But, then, $u_k(C) > u_k(C')$, contradicting condition (8).

If $u_i(C) = -\lambda_1 - 2\lambda_2$, then by Corollary 2, there are exactly two vertices v_i and v_l in $\tilde{N}_i$ that are gray in C and the other vertices in $N_i \cup N_l$ are all white in C . By condition (8), one has $u_i(C') \geq -\lambda_1 - 2\lambda_2$. Then, there must exist $v_k \in N_i^2(C') \cup N_l^2(C')$, otherwise $u_i(C') \leq -2\lambda_2 m_i(C') - \lambda_2 m_l(C') = -3\lambda_2 < u_i(C)$. If $v_k = v_l$, then v_k is gray in C , and $u_k(C) = -\lambda_1 - 2\lambda_2$. If $v_k \neq v_l$, then v_k is white in C , and $u_k(C) \geq -\lambda_2(m_i(C) + m_l(C)) = -2\lambda_2$ by (10). In any case, $u_k(C) > u_k(C') = -4\lambda_1$, a contradiction.

Claim 3: For any v_i with $c_i = 0$, one has $c'_i = 0$.

If the claim is not true, then by Claim 1, one has $c'_i = 1$. By Corollary 1 (ii), one has $u_i(C) = -|N_i^1(C)|\lambda_2$. For any gray vertex $v_k \in N_i^1(C)$, by Claim 2, v_k is also gray in C' . Since C is an NE, by Corollary 1 (i), $N_k^2(C) = \emptyset$, and thus by Claim 1, $N_k^2(C') = \emptyset$. Hence $m_k(C') = 1$, and thus $u_i(C') \leq -\lambda_1 - |N_i^1(C)|\lambda_2$. Consequently, $u_i(C') - u_i(C) \leq -\lambda_1 < 0$, a contradiction.

Claim 4: For any v_i with $c_i = 2$, one has $c'_i = 2$.

Suppose $c'_i \neq 2$. Because C is an M-RDF by Theorem 2, strategy profile $(c'_i \neq 2, C_{-i})$ is not an RDF. So, there is a white vertex $v_l \in N_i^0(C)$ with $N_l^2(c'_i \neq 2, C_{-i}) = \emptyset$. By Claim 2 and Claim 3, any vertex $v_k \in \tilde{N}_l$ with $k \neq i$ has $c'_k = c_k$. So, $N_l^2(C') = \emptyset$, and thus $m_l(C') = 1$, $u_l(C') = -2\lambda_2 - \lambda_2 \sum_{v_k \in N_l} (2 - c_k)m_k(C')$ and $u_l(C) = -\lambda_2 \sum_{v_k \in N_l} (2 - c_k)m_k(C)$. Note that $m_k(C) \leq m_k(C')$ (if $m_k(C) = 1$. Therefore, $\tilde{N}_k^2(C) = \emptyset$, and by Claim 2 and Claim 3, one has $\tilde{N}_k^2(C') = \emptyset$, which implies that $m_k(C') = 1$). Then, $u_l(C') - u_l(C) \leq -2\lambda_2 < 0$, a contradiction.

Combining the above leads to a contradiction where $C' = C$. ■

IV. ALGORITHM DESIGN AND ANALYSIS

In this section, three algorithms will be presented: Game-based asynchronous algorithm (GAA), game-based synchronous algorithm (GSA) and enhanced game-based synchronous algorithm (EGSA). GAA is a direct simulation of the RDG. It converges to an NE in $O(n)$ rounds of interactions, and the decision of each player depends only on local information in its neighborhood. However, GAA is a sequential algorithm. GSA improves the efficiency of GAA by a distributed realization of the game, and thus can be better imple-

mented by multi-agent systems. EGSA further extends GSA with the concept of a private contract, which is inspired by a cooperative game. It will be shown that EGSA can achieve a better solution than GSA.

A. Game-Based Asynchronous Algorithm

The pseudo code of GAA is presented in Algorithm 1. It is a naive simulation of the RDG. By the definition of utility functions, each player can make their decision locally: suppose each player v_j stores $SS_j = (c_j, m_j(C))$ with respect to the current solution C . Then, each player v_i can decide on his best response $BR(v_i, C)$ based on $\{SS_j\}_{v_j \in \bar{N}_i}$. In the algorithm, parameter $T = \frac{(4\lambda_1 + 2\lambda_2)n}{\min\{3\lambda_1 - 2\lambda_2, -4\lambda_1 + 3\lambda_2\}}$. By Theorem 5, an NE can be reached in at most this number of iterations.

Algorithm 1 Game-based Asynchronous Algorithm (GAA)

Input: An initial strategy profile $C^{(0)} = (c_1^{(0)}, \dots, c_n^{(0)})$
Output: An NE C'

- 1: $C \leftarrow C^{(0)}$.
- 2: **for** $t = 1, 2, \dots, T$ **do**
- 3: **for** $i = 1, 2, \dots, n$ **do**
- 4: $c'_i \leftarrow BR(v_i, C)$ by accessing SS_j for $v_j \in \bar{N}_i$
- 5: **if** $u_i(c'_i, C_{-i}) > u_i(c_i, C_{-i})$ **then**
- 6: $c_i \leftarrow c'_i$
- 7: **end if**
- 8: **end for**
- 9: **if** $C = C^{(t-1)}$ **then**
- 10: Break and go to line 15
- 11: **else**
- 12: $C^{(t)} \leftarrow C$
- 13: **end if**
- 14: **end for**
- 15: Output $C' \leftarrow C$

Note that players in GAA make decisions in a certain order. So, GAA is a sequential algorithm. To realize the game in a distributed manner, a natural idea is to let the players make decisions simultaneously. However, such a method may induce chaos to prevent the algorithm from converging. Consider the example in Fig. 7. Suppose the current strategy profile is $C = (0, 0, 0, 0)$. For any player v_i , since $u_i(c'_i = 2, C_{-i}) = -4\lambda_1 > -\lambda_1 - 5\lambda_2 = u_i(c''_i = 1, C_{-i}) > -6\lambda_2 = u_i(C)$, the best response of v_i is to change 0 to 2. If all players take their best responses simultaneously, then the next strategy profile is $C' = (2, 2, 2, 2)$. Now, for any player v_i , since $u_i(c'_i = 0, C_{-i}) = 0 > -\lambda_1 = u_i(c''_i = 1, C_{-i}) > -4\lambda_1 = u_i(C)$, his best response is to change from 2 to 0. A simultaneous action makes the strategy profile back to $C = (0, 0, 0, 0)$.

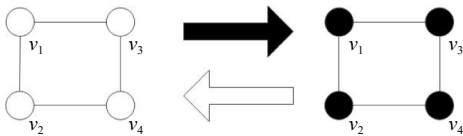


Fig. 7. Simultaneous update may result in an infinite loop.

To avoid such a mess, we restrict simultaneous changes to be made by a set of independent players, which leads to the GSA to be discussed in the next subsection.

B. Game-Based Synchronous Algorithm

The pseudo code of GSA is presented in Algorithm 2. The parameter T is also $\frac{(4\lambda_1 + 2\lambda_2)n}{\min\{3\lambda_1 - 2\lambda_2, -4\lambda_1 + 3\lambda_2\}}$, the rationale of which is supported by Lemma 7. For the current strategy profile C , denote by $mu_i(C) = u_i(BR(v_i, C), C_{-i}) - u_i(c_i, C_{-i})$ the marginal utility of player v_i . In each round of iteration, all players compute their marginal utilities simultaneously, but not all those players with positive marginal utilities take actions. A player v_i decides to change his strategy only when

$$i = \arg \min \{j : v_j \in \bar{N}_{i,2} \text{ and } mu_j(C) > 0\}. \quad (11)$$

Algorithm 2 Game-based Synchronous Algorithm (GSA)

Input: An initial strategy profile $C^{(0)} = (c_1^{(0)}, \dots, c_n^{(0)})$
Output: An NE C'

- 1: $C \leftarrow C^{(0)}$.
- 2: **for** $t = 1, 2, \dots, T$ **do**
- 3: **for every** player v_i (this is done simultaneously) **do**
- 4: $c'_i \leftarrow BR(v_i, C)$ by accessing SS_j for $v_j \in \bar{N}_i$
- 5: $mu_i \leftarrow u_i(c'_i, C_{-i}) - u_i(c_i, C_{-i})$
- 6: **if** v_i satisfies equation (11) **then**
- 7: $c_i \leftarrow c'_i$
- 8: **end if**
- 9: **end for**
- 10: **if** $C = C^{(t-1)}$ **then**
- 11: Break and go to line 15
- 12: **else**
- 13: $C^{(t)} \leftarrow C$
- 14: **end if**
- 15: **end for**
- 16: Output $C' \leftarrow C$

That is, player v_i has the priority to change his strategy only when he has the smallest ID among those players in $\bar{N}_{i,2}$ with positive marginal utility.

The following observation shows that those players who actually change their strategies form an independent set.

Observation 2: Suppose $\mathcal{I}$ is the set of players who have actually changed their strategies simultaneously in a round of the parfor loop of Algorithm 2. Then, $\mathcal{I}$ is an independent set in the following sense:

$$\bar{N}_i \cap \bar{N}_j = \emptyset, \quad \forall v_i, v_j \in \mathcal{I}.$$

In fact, if $\bar{N}_i \cap \bar{N}_j \neq \emptyset$, then $d(i, j) \leq 2$, and thus $v_i \in \bar{N}_{j,2}$ and $v_j \in \bar{N}_{i,2}$. By the rule in (11), for v_i and v_j , only the one with the smaller ID can belong to $\mathcal{I}$.

Intuitively, since every v_i makes his decision depending only on $\bar{N}_i(C)$, if no other players in $\bar{N}_i$ change their strategies at the same time, then the decision made by v_i stays as the best response for the altered strategy profile. Thus, allowing players in an independent set to change strategies simultaneously can effectively decouple mutual influences. A more detailed analysis is given in the following.

Lemma 7: Suppose C is the current strategy profile. After one round of the parfor loop of Algorithm 2, the new strategy profile is $C' = (C'_\mathcal{I}, C_{-\mathcal{I}})$, where $\mathcal{I}$ is the set of players who have strategies changed in this round. Then,

$$\pi(C') - \pi(C) = \sum_{v_i \in \mathcal{I}} (u_i(c'_i, C_{-i}) - u_i(c_i, C_{-i})).$$

Proof: Suppose $\mathcal{I} = \{v_{i_1}, v_{i_2}, \dots, v_{i_t}\}$. By Observation 2, the player set V can be decomposed to a disjoint union of sets $V = \bar{N}_{i_1} \cup \bar{N}_{i_2} \cup \dots \cup \bar{N}_{i_t} \cup V_r$, where $V_r = V \setminus \bigcup_{l=1}^t \bar{N}_{i_l}$. Then, the potential functions $\pi(C)$ and $\pi(C')$ can be rewritten as

$$\begin{aligned} \pi(C) &= -\lambda_1 \sum_{v_i \in \mathcal{I}} c_i^2 - \lambda_2 \sum_{v_i \in \mathcal{I}} \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C) \\ &\quad - \lambda_1 \sum_{v_i \notin \mathcal{I}} c_i^2 - \lambda_2 \sum_{v_j \in V_r} (2 - c_j) m_j(C) \\ \pi(C') &= -\lambda_1 \sum_{v_i \in \mathcal{I}} c_i'^2 - \lambda_2 \sum_{v_i \in \mathcal{I}} \sum_{v_j \in \bar{N}_i} (2 - c'_j) m_j(C') \\ &\quad - \lambda_1 \sum_{v_i \notin \mathcal{I}} c_i'^2 - \lambda_2 \sum_{v_j \in V_r} (2 - c'_j) m_j(C'). \end{aligned} \quad (12)$$

Note that $c'_i = c_i$ for any $v_i \notin \mathcal{I}$. For any $v_i \in \mathcal{I}$ and any $v_j \in \bar{N}_i$, by the independence of $\mathcal{I}$, v_i is the only vertex in $\bar{N}_j$ whose c -value is changed. Since the value of m_j is determined by the c -values in $\bar{N}_j$, one has $m_j(C') = m_j(c'_i, C_{-i})$. For any $v_j \in V_r$, one has $\bar{N}_j \cap \mathcal{I} = \emptyset$. So $m_j(C') = m_j(C)$. Thus, $\pi(C')$ can be written as

$$\begin{aligned} \pi(C') &= -\lambda_1 \sum_{v_i \in \mathcal{I}} c_i'^2 - \lambda_1 \sum_{v_i \notin \mathcal{I}} c_i^2 - \lambda_2 \sum_{v_j \in V_r} (2 - c_j) m_j(C) \\ &\quad - \lambda_2 \sum_{v_i \in \mathcal{I}} \left((2 - c'_i) m_i(c'_i, C_{-i}) + \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(c'_i, C_{-i}) \right). \end{aligned} \quad (13)$$

The lemma follows by subtracting (12) from (13). ■

Theorem 7: Starting from any initial strategy profile $C^{(0)}$, the number of rounds for GSA 2 to reach an NE is at most $\frac{(4\lambda_1 + 2\lambda_2)n}{\min\{3\lambda_1 - 2\lambda_2, -4\lambda_1 + 3\lambda_2\}} = O(n)$.

Proof: Observe that player v_i is willing to change his strategy only when $mu_i(C) > 0$. By Lemma 7, $\pi(C') - \pi(C) = \sum_{v_i \in \mathcal{I}} (u_i(c'_i, C_{-i}) - u_i(c_i, C_{-i})) \geq \sum_{v_i \in \mathcal{I}} (u_i(c'_i, C_{-i}) - u_i(c_i, C_{-i})) = mu_i(C)$. The remaining proofs are similar to that of Theorem 5. ■

C. Enhanced Game-Based Synchronous Algorithm

Although GSA can obtain an NE in a linear number of rounds of iterations, it is an S-RDF by Corollary 3, but the gap between an NE and a global optimal solution might be large. Consider the example in Fig. 8, where C is an NE with $\gamma_R(G, C) = n - 1$, and C' is a G-RDF with $\gamma_R(G, C') = 2$.

There are two possible reasons for the above problems:

- 1) The range of information that a player uses is small;
- 2) A player is assumed to be selfish, i.e., a player only maximizes his own utility regardless of the other players.

To avoid getting stuck in a bad NE such as the one in Fig. 8(a), the players should be more cooperative, that is, there is a coalition that can make decisions simultaneously by cooperating. A large-scale coalition is intractable, a local coalition might be good enough in terms of both performance and computational complexity. For this reason, a concept of private contract is proposed as follows, which borrows the idea of

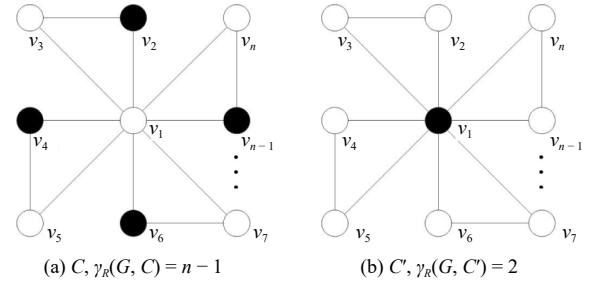


Fig. 8. An example showing that the gap between an NE and a G-RDF can be very large.

contract from cooperative game theory.

Definition 10 (Private contract): In the current strategy profile C , a private contract is proposed by a player v_i which suggests that all players v_j in a coalition $\tau_i \subseteq V$ (with $v_i \in \tau_i$) change their strategies from c_j to c'_j . A private contract is *valid* if all players in τ_i agree with it, and the next strategy profile becomes $(C'_{\tau_i}, C_{-\tau_i})$.

In the following, a private contract $\mathcal{A}$ is used, which is proposed by v_i , suggesting $c'_i = 2$ and $c'_j = 0$ for any $v_j \in \tau_i \setminus \{v_i\}$, where $\tau_i \setminus \{v_i\} = N_i^1(C) \cup \{v_j \in N_i^2(C) : m_{i,j}(C) = 1\}$, with

$$m_{i,j}(C) = \begin{cases} 1, & \bar{N}_j(C) \subseteq N_i \\ 0, & \text{otherwise} \end{cases} \quad (14)$$

and $\bar{N}_j(C) = \{v_k \in N_j^0(C) : N_k^2(C) = \{v_j\}\}$.

Given a private contract proposed by v_i , assume that all players in coalition τ_i will agree with this contract if

$$\Delta_u(C'_{\tau_i}, C) = \sum_{v_j \in \tau_i} (u_j(C'_{\tau_i}, C_{-\tau_i}) - u_j(C)) > 0. \quad (15)$$

The ideal is that strictly positive utility gain $\Delta_u(C'_{\tau_i}, C)$ can be reasonably distributed among the coalition so that each player in τ_i can get a strictly positive utility gain. Therefore, such an assumption is reasonable.

The enhanced algorithm EGSA is described in Algorithm 3, where $w_i(C)$ in line 12 of the algorithm is defined as follow:

$$w_i(C) = \sum_{v_j \in N_i^1(C)} m_j(C) + 2 \sum_{v_j \in N_i^2(C)} m_{i,j}(C).$$

It will be proved that if the condition in line 12 is satisfied, then the contract proposed by v_i can always be agreed. The algorithm EGSA first produces an NE C , which can be recognized by the criterion $C = C^{(t-1)}$. If a player v_i proposes a private contract $\mathcal{A}$ that is agreed by the coalition, then C is changed to $C' = (C'_{\tau_i}, C_{-\tau_i})$. The algorithm continues to produce an NE from C' . This process is repeated until no player proposes new private contract (i.e. no player satisfies the condition in line 12 of EGSA). At the termination, the output is an NE.

The following lemma describes a property that assists in proving a condition for contract $\mathcal{A}$ to be valid as well as in analyzing the time complexity of EGSA.

Lemma 8: In a strategy profile C , for any player v_i with $c_i = 0$ and for any player $v_j \in N_i^2(C)$, if $m_{i,j}(C) = 1$, then $C' = (c'_i = 2, c'_j = 0, C_{-\{i,j\}})$ will not create any new white free vertex.

Proof: If v_j changes his strategy from 2 to 0, then all new white free vertices, if any, must belong to $\tilde{N}_j(C)$. Because $m_{i,j}(C) = 1$, one has $\tilde{N}_j(C) \subseteq N_i$. Since v_i changes his strategy from 0 to 2, it can strongly dominate all white vertices in $N_i \supseteq \tilde{N}_j(C)$. ■

Algorithm 3 Enhanced Game-based Synchronous Algorithm (EGSA)

Input: An initial strategy profile $C^{(0)} = (c_1^{(0)}, \dots, c_n^{(0)})$
Output: An NE C'

```

1:  $C \leftarrow C^{(0)}$ .
2: for  $t = 1, 2, \dots, T$  do
3:   for every player  $v_i$  (this is done simultaneously) do
4:      $c'_i \leftarrow BR(v_i, C)$  by accessing  $SS_j$  for  $v_j \in \tilde{N}_i$ 
5:      $mu_i \leftarrow u_i(c'_i, C_{-i}) - u_i(c_i, C_{-i})$ 
6:     if  $v_i$  satisfies (11) then
7:        $c_i \leftarrow c'_i$ 
8:     end if
9:   end for
10:  if  $C = C^{(t-1)}$  then
11:    for  $i = 1, 2, \dots, n$  do
12:      if  $c_i = 0$  and  $w_i(C) \geq 3$  then
13:         $v_i$  proposes a private contract  $\mathcal{A}$ 
14:         $C \leftarrow (C'_{\tau_i}, C_{-\tau_i})$  (every player  $v_j \in \tau_i$  agree with  $\mathcal{A}$  by Lemma 9)
15:         $C^{(t)} \leftarrow C$ 
16:        Break and go to line 21
17:      end if
18:    end for
19:  else
20:     $C^{(t)} \leftarrow C$ 
21:  end if
22: end for
23: Output:  $C' \leftarrow C$ 

```

The next lemma gives a sufficient condition for a private contract to be agreed by the intended coalition.

Lemma 9: For any NE C , the contract $\mathcal{A}$ proposed by player v_i with $c_i = 0$ and $w_i(C) \geq 3$ is valid.

Proof: To prove the lemma, it is needed to prove (15). Let $x_1 = |N_i^1(C)|$ and $x_2 = |\{v_j \in N_i^2(C) : m_{i,j}(C) = 1\}|$.

Because C is an NE, by Corollary 1, every gray vertex v_j is not adjacent to any black vertex, and thus $m_j(C) = 1$; and every white vertex v_k is adjacent to at least one black vertex, and thus $m_k(C) = 0$. So, $u_i(C) = -\lambda_2 x_1$. By Lemma 1, $u_i(C'_{\tau_i}, C_{-\tau_i}) = -4\lambda_1$. Hence, $u_i(C'_{\tau_i}, C_{-\tau_i}) - u_i(C) = \lambda_2 x_1 - 4\lambda_1$ and $w_i(C) = x_1 + 2x_2$.

For any vertex $v_j \in N_i^1(C)$, with similar reasons as the above, one obtains $u_j(C) = -\lambda_1 - \lambda_2 - \lambda_2 |N_j^1(C)|$. By Lemma 8, there is no white free vertex produced in $C' = (C'_{\tau_i}, C_{-\tau_i})$, then $u_j(C') = -\lambda_2 |N_j^1(C')|$. Note that $N_j^1(C') \subseteq N_j^1(C)$. So, $u_j(C') \geq -\lambda_2 |N_j^1(C)|$, and thus $u_j(C') - u_j(C) \geq \lambda_1 + \lambda_2 > 0$.

For any vertex $v_k \in N_i^2(C)$ with $m_{i,k}(C) = 1$, by Corollary 1, one has $N_k^1(C) = \emptyset$. Note that gray vertices in C stays as gray in C' . Hence, $N_k^1(C') = \emptyset$. So, $u_k(C') = 0$ by Lemma 8 and $u_k(C) = -4\lambda_1$ by Lemma 1. Thus, $u_k(C') - u_k(C) = 4\lambda_1 > 0$

Consequently, $\sum_{v_j \in \tau_i} (u_j(C'_{\tau_i}, C_{-\tau_i}) - u_j(C)) \geq -4\lambda_1 + \lambda_2 x_1 +$

$(\lambda_1 + \lambda_2)x_1 + 4\lambda_1 x_2 > 2\lambda_1(x_1 + 2x_2) + \lambda_2 x_1 - 4\lambda_1$. Because $w_i(C) \geq 3$ means $x_1 + 2x_2 \geq 3$, $\sum_{v_j \in \tau_i} (u_j(C'_{\tau_i}, C_{-\tau_i}) - u_j(C)) \geq 6\lambda_1 + \lambda_2 x_1 - 4\lambda_1 = 2\lambda_1 + \lambda_2 x_1 > 0$. The contract is valid. ■

Next, the objective is to determine the convergence time of EGSA. For this purpose, two lemmas will be established first. The idea is as follows. When the algorithm finds an NE C , if a private contract proposed by v_i is agreed, then the strategy profile becomes $C' = (C'_{\tau_i}, C_{-\tau_i})$. Lemma 10 shows that $\gamma_R(G, C') < \gamma_R(G, C)$. Because γ_R can only take integer values, the γ_R -value is *strictly* decreased by at least 1. Note that except for the private contract part, EGSA is the same as GSA. Hence, by Theorem 7, starting from C' , the algorithm reaches an NE C'' in $O(n)$ rounds. Lemma 11 implies that $\gamma_R(G, C'') \leq \gamma_R(G, C')$. So, in the whole process, the γ_R -value is monotone non-decreasing, and in at most $O(n)$ rounds, the γ_R -value is decreased by at least 1. Since the γ_R -value is upper bounded by $2n$, the algorithm terminates in $O(n^2)$ rounds. Next, proofs are given to verify these results.

Lemma 10: Let C be an NE. If a player v_i proposes a valid private contract, then the resulting strategy profile $C' = (C'_{\tau_i}, C_{-\tau_i})$ is an RDF and $\gamma_R(G, C') < \gamma_R(G, C)$.

Proof: Since C is an NE, by Theorem 1, it is an RDF, and thus there is no free white vertex in C . Note that any vertex $v_j \in \tau_i$ with $c_j = 1$ is dominated by v_i in C' (because $c'_i = 2$), and changing $c_j = 1$ to $c'_j = 0$ does not affect a white vertex to be strongly dominated or not (by Observation 1). Furthermore, by Lemma 8, changing the strategy of a vertex $v_j \in \tau_i$ with $c_j = 2$ will not create a new free white vertex. Hence, there is no free white vertex in C' either, and thus C' is also an RDF.

Note that a player v_i will propose a private contract only when $w_i(C) \geq 3$. Let x_1 and x_2 be the numbers of gray vertices and black vertices in τ_i , respectively. Because $w_i(C) \geq 3$ implies $x_1 + 2x_2 \geq 3$, one has $\gamma_R(G, C') = \gamma_R(G, C) + 2 - x_1 - 2x_2 \leq \gamma_R(G, C) - 1$. ■

Lemma 11: Starting from any RDF C , let C' be the first NE reached by the algorithm after C . Then, $\gamma_R(G, C') \leq \gamma_R(G, C)$.

Proof: Let x_{ij} be the number of players changing their strategies from i to j during the iterations from C to C' . It will be proved that

$$x_{01} + x_{12} + 2x_{02} - x_{10} - x_{21} - 2x_{20} \leq 0 \quad (16)$$

which implies the monotonicity of γ_R . Details are provided in the Appendix B. ■

Call the output of EGSA as an enhanced Nash equilibrium (ENE). As commented before, it is indeed an NE. Furthermore, as Lemmas 10 and 11 imply, the size of an ENE is smaller than the size of the NE output by GSA, as long as some private contract is proposed. The time complexity of EGSA is given below, which follows from Lemmas 10 and 11 and the argument before their proofs.

Theorem 8: Starting from any initial state $C^{(0)}$, the number of rounds for EGSA to converge to an ENE is $O(n^2)$.

Let S_{ENE} be the set of solutions that might be output by EGSA. As the example in Fig. 8 indicates, not all NE belong to S_{ENE} . Furthermore, every G-RDF is an ENE. In fact, Theorem 4 implies that any G-RDF is an NE. If it is not an ENE, then Lemma 10 indicates that its γ -value can be strictly decreased, contradicting the definition of G-RDF. Combining

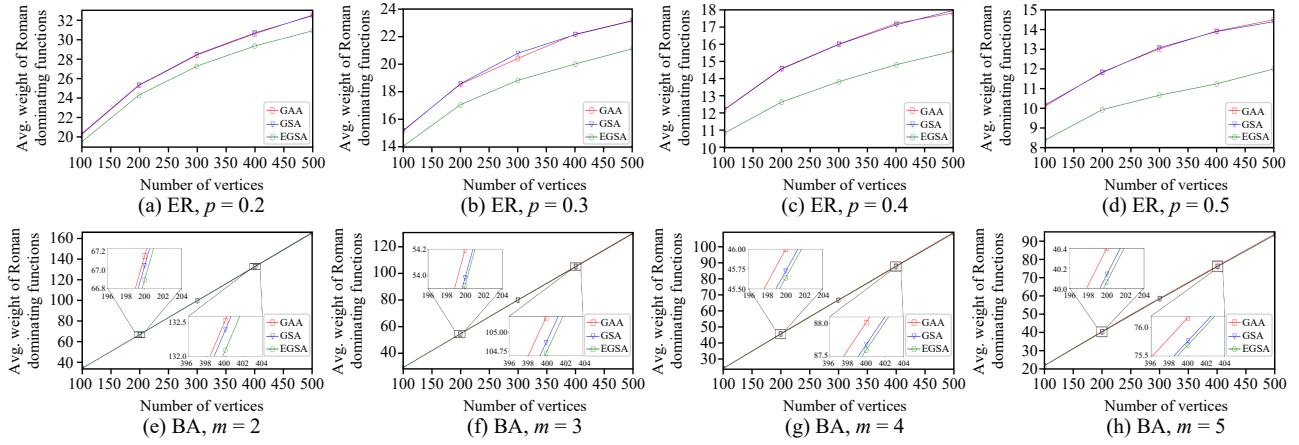


Fig. 9. Comparison of GAA, GSA and EGSA.

these with Corollary 3, the following relations are revealed, and the relationship among these sets is illustrated in Fig. 5.

Corollary 4: In an RDG, $S_{G-RDF} \subseteq S_{ENE} \subseteq S_{NE} \subseteq S_{S-RDF} \subseteq S_{M-RDF} \subseteq S_{RDF}$.

V. SIMULATION RESULTS

This section reports on the experiments on the performances of the algorithms GAA, GSA and EGSA. All experiments are coded in Python and run with an identical configuration: AMD Ryzen 5 3500U with Radeon Vega Mobile Gfx and 16 GB of RAM.

A. Comparing GAA, GSA and EGSA

In this section, the three algorithms GAA, GSA and EGSA are compared in terms of accuracy and time complexity. Graphs for the experiments are generated randomly using the following two models.

1) *The Barabási-Albert Graph (BA)* [37]: Starting from a graph with a small number m_0 of vertices, new vertices are iteratively added. When a new vertex is added, it is connected to m existing vertices, where $m \leq m_0$, and the probability that an existing vertex is linked with the new vertex is proportional to its current degree.

2) *The Erdős-Rényi Graph (ER)* [38]: In this graph, every pair of vertexes are connected by an edge with probability p .

In Fig. 9, the horizontal axis is the number of vertices n and the vertical axis shows the average weight of the solutions obtained on 1000 randomly sampled graphs. It can be seen that EGSA is generally better than GSA. Its advantage is comparatively more obvious on ER graphs, but not so obvious on BA graphs. Maybe because BA graphs have good structures so that GSA already performs fairly well on them. Note that the performance of GSA is similar to that of GAA in terms of the weights of the solutions. This is because GSA is essentially a distributed realization of GAA. For ER graphs with different p -values, the weights of computed solutions decrease with the increase of p -values. This is natural, as denser graphs provide more links for domination. In any case, the advantage of EGSA over GSA can be witnessed. A similar phenomenon is observed on BA graphs with varying m . So for clarity, the following experiments only show the results of $m = 5$ for BA graphs and $p = 0.2$ for ER graphs.

Table III shows the average number of rounds of the three algorithms, where one round refers to one iteration of the *outer for loop*. Although it seems that the number of rounds of GAA is smaller than that of GSA and EGSA, it should be noted that in each round of GAA, a centralized controller has to compute n players' best responses sequentially, while in each round of GSA, all players compute their best responses simultaneously. Therefore, the real time for GAA is n times the number of rounds while the real time for GSA is just the number of rounds. Let n_{GAA} and n_{GSA} be the number of rounds in GAA and GSA, respectively. Define $\eta = \frac{n_{GSA}}{n_{GAA} \times n}$ and use it to measure the ratio in real time. The values of η are shown in Table IV. It can be seen that, in terms of real time, GSA is much faster than GAA. Furthermore, it can be observed that the acceleration effect is more prominent with the increase of n , especially on ER.

TABLE III
AVERAGE ROUNDS OF GAA, GSA AND EGSA

Graph	Algorithm	$ V = 100$	200	300	400	500
BA	GAA	2.332	2.491	2.615	2.744	2.879
	GSA	17.968	30.278	41.199	50.679	59.314
	EGSA	18.093	30.372	41.296	50.777	59.391
ER	GAA	2.846	3.062	3.111	3.190	3.152
	GSA	22.309	27.267	29.520	31.369	32.287
	EGSA	23.364	28.586	31.031	33.030	34.241

TABLE IV
COMPARE TIME COMPLEXITY OF GA AND
GSA MEASURED BY η (%)

Graph	$ V = 100$	200	300	400	500
BA	7.70	6.08	5.25	4.62	4.12
ER	7.84	4.45	3.16	2.46	2.05

B. Comparing GSA and Greedy Algorithm

Since the MinRD problem is NP-hard, one cannot expect a polynomial-time algorithm to obtain the exact solution. The best-known approximation algorithm for MinRD is a greedy algorithm (GA) proposed in [39], which can achieve asymp-

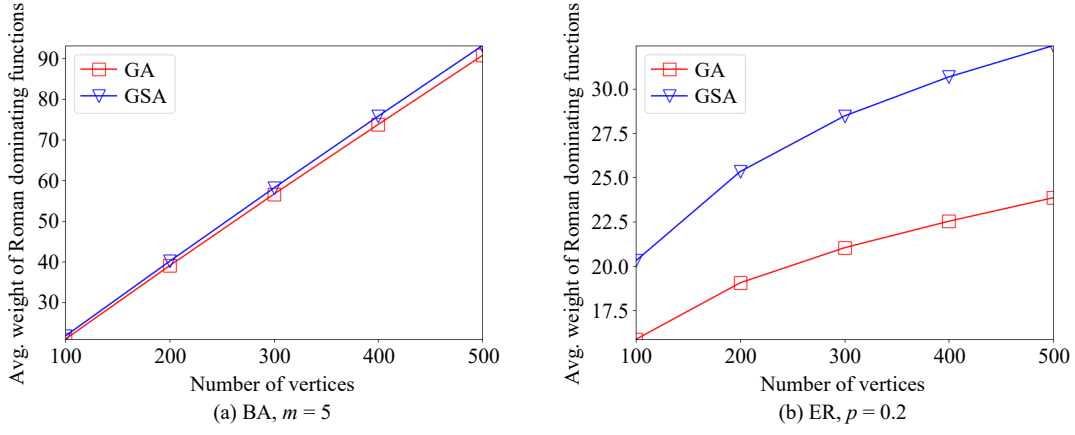


Fig. 10. Comparison of GA and GSA.

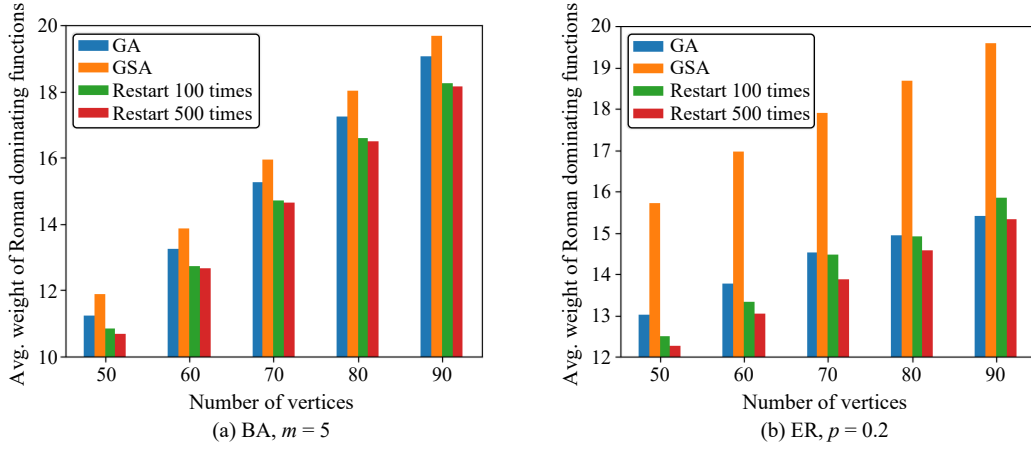


Fig. 11. Compare GA and GSA using restart.

totically a tight logarithmic approximation ratio.

To compare GSA with GA on BA and ER, Fig. 10 shows the average weight of solutions computed by GSA (line with triangles) and GA (line with rectangles). In terms of weight, GA is smaller than GSA, especially on ER. This is reasonable, because GA is a centralized algorithm. The advantage of GSA is that it is a distributed algorithm, which is more preferred in large-scale networks. Furthermore, for the experiment in Fig. 10, GSA always starts from the initial strategy profile $(0, \dots, 0)$. A question is: if the algorithm is restarted from different initial strategies, will it perform better? Note that Theorem 7 guarantees that GSA can converge to an NE from *any* initial strategy profile. Fig. 11 shows the average weight by restarting 100 times and 500 times (also tested for restarting 200, 300 and 400 times). It can be seen that restarting GSA can achieve better solutions than GA. Note that GA is a deterministic algorithm, which cannot be benefited from restarting. It has also been tested and noticed that the average weight will not be significantly improved with more than 100 times.

C. Comparing With an Exact Solution on a Tree

To see the accuracy of the three proposed algorithms, they are compared with the exact algorithm (DP) for MinRD on trees, which is a dynamic program proposed in [6]. Trees for the experiments are generated in two ways as follows.

1) *Barabási-Albert Tree (BAT)*: A tree is constructed by the

BA model with $m_0 = 2$ and $m = 1$.

2) *Random Tree (RT)*: Let V be a vertex set on n vertices and F be the edge set consisting of all possible edges between vertices of V . Starting from an empty graph formed from vertex set V , iteratively add an edge e from F randomly and uniformly as long as no cycle is created, until a spanning tree on V is obtained.

For each n , 1000 trees of size n are sampled randomly. Table V shows the relative error $\omega = \frac{\gamma_R(G, f) - \gamma_R(G, f^*)}{\gamma_R(G, f^*)}$ for the RDF f obtained by the algorithms (GA, GSA and EGSA) and the optimal solution f^* generated by DP. Here, the results of GAA are not presented since GSA has the same accuracy as GAA, but the result of the greedy algorithm GA is included for a comparison. As can be seen from Table V, GSA is superior to GA on RT. Although GA is better than GSA on BAT, with the restarting strategy, GSA is much better than GA. Furthermore, EGSA performs fairly well on trees, yielding solutions very close to an optimal one, the gap is less than 0.01%. Note that EGSA outperforms GSA even if GAS restarts 100 times.

VI. CONCLUSION

In this paper, we study the minimum Roman domination problem (MinRD) in a multi-agent system by game theory. A Roman domination game is proposed and the existence of a Nash equilibrium (NE) is guaranteed by proving that the game

TABLE V
RELATIVE ERRORS OF THE PROPOSED ALGORITHMS AND
DP MEASURED BY ω (%)

Graph	Algorithm	$ V = 100$	200	300	400	500
RT	GA	1.81	2.10	1.87	1.82	1.82
	GSA	1.09	1.26	1.15	1.15	1.16
	GSA(100)	0.24	0.41	0.60	0.80	0.92
	EGSA	0.12	0.37	0.31	0.33	0.32
BAT	GA	0.17	0.14	0.19	0.19	0.18
	GSA	0.41	0.35	0.40	0.37	0.36
	GSA(100)	0.002	0.004	0.01	0.02	0.02
	EGSA	0.01	0.01	0.01	0.02	0.004

is a potential game. Furthermore, we classify three types of RDF as S_{M-RDF} , S_{S-RDF} and S_{G-RDF} ; we show that $S_{G-RDF} \subseteq S_{ENE} \subseteq S_{NE} \subseteq S_{S-RDF} \subseteq S_{M-RDF} \subseteq S_{RDF}$, and prove that every NE is a Pareto-optimal solution. In addition, three algorithms for RDG are proposed, named GAA, GSA and EGSA. In all three algorithms, players only use local information to make decisions, which greatly improves the efficiency of the algorithms. In GAA, an NE can be found in a linear number of rounds of interactions. While GSA can not only achieve the same performance as GAA, its computation can also be realized distributedly, which is more preferred in a large-scale network. In EGSA, an enhanced NE can be obtained, which can further improve the quality of the solution. These theoretical results are supported by experiments.

In the future, we will explore different game models on domination problems, such as allocation games [40], evolutionary games [41] and networked games [42], [43], hoping to obtain improved performance. Another direction worthy of exploration is to incorporate Reinforcement Learning in the study. There are also many interesting topics to be further explored, such as Price of Anarchy (PoA) [44], core of coalitional games and the games with imperfect information [45].

APPENDIX A

DETAILED PROOFS FOR TABLE II

In the proof of Theorem 5, we have shown a positive lower bound for Δu_i in the case “ $c_i = 2, c'_i = 0$ ”. In the following, the detailed analysis for the other cases are provided.

If $c_i = 2$ and $c'_i = 1$, $u_i(C) = -4\lambda_1$ by Lemma 1 and $u_i(c'_i = 1, C_{-i}) = -\lambda_1 - \lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C')$. Because $3\lambda_1 - 2\lambda_2 > 0$ and $3\lambda_1 - 3\lambda_2 < 0$, the positive value of $u_i(c'_i = 1, C_{-i}) - u_i(C)$ in this case must be $\geq 3\lambda_1 - 2\lambda_2$.

If $c_i = 1$ and $c'_i = 0$, $u_i(C) = -\lambda_1 - \lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C)$ and $u_i(c'_i = 0, C_{-i}) = -\lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C')$. Because $\lambda_1 > 0$ and $\lambda_1 - \lambda_2 < 0$, the positive value of $u_i(c'_i = 0, C_{-i}) - u_i(C)$ in this case must be $\geq \lambda_1$.

If $c_i = 0$ and $c'_i = 1$, $u_i(C) = -\lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C)$ and $u_i(c'_i = 1, C_{-i}) = -\lambda_1 - \lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C')$. Because $-\lambda_1 + \lambda_2 > 0$ and $-\lambda_1 < 0$, the positive value of $u_i(c'_i = 1, C_{-i}) - u_i(C)$ in this case must be $\geq -\lambda_1 + \lambda_2$.

If $c_i = 1$ and $c'_i = 2$, $u_i(C) = -\lambda_1 - \lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C)$ and $u_i(c'_i = 2, C_{-i}) = -4\lambda_1$. Because $-3\lambda_1 + 3\lambda_2 > 0$ and $-3\lambda_1 +$

$2\lambda_2 < 0$, the positive value of $u_i(c'_i = 2, C_{-i}) - u_i(C)$ in this case must be $\geq -3\lambda_1 + 3\lambda_2$.

If $c_i = 0$ and $c'_i = 2$, $u_i(C) = -\lambda_2 \sum_{v_j \in \bar{N}_i} (2 - c_j) m_j(C)$ and $u_i(c'_i = 2, C_{-i}) = -4\lambda_1$. Because $-4\lambda_1 + 3\lambda_2 > 0$ and $-3\lambda_1 + 2\lambda_2 < 0$, the positive value of $u_i(c'_i = 2, C_{-i}) - u_i(C)$ in this case must be $\geq -4\lambda_1 + 3\lambda_2$.

APPENDIX B

PROOF OF LEMMA 11

Lemma 11 is proved by a series of lemmas below. The idea of the proofs is as follows. In both C and C' , there is no white free vertex. It is needed to estimate, in various cases, how many white free vertices are created and diminished, resulting in an upper bound for the increase of the number of white free vertices, which is then compared with 0. Since C' is an NE, every gray vertex is free. Denote by y the number of gray vertices in C that are strongly dominated. Then, they will diminish in C' . Hence, when reaching C' , the total number of gray vertices that are strongly dominated is decreased by y . Motivated by this observation, it will then be needed to estimate the number of created and diminished gray vertices that are strongly dominated during the evolution from C to C' , which will be compared to the total decrease with y . Using these relations, by some algebraic manipulation, (16) can be obtained, implying the monotonicity of γ .

In the following, the number of changes after a strategy profile C is estimated and then changed to C' . Note that, when a player v_i changes his strategy, it only affects the statuses of those players in $\bar{N}_i$ (to become free or become strongly dominated). So, it is only needed to consider those vertices in $\bar{N}_i$.

Lemma 12: In any strategy profile C , when a player v_i changes his strategy from 2 to 0, one of the following two situations holds.

- 1) At most one free white vertex is created and no gray vertex becomes free;
- 2) At most two gray vertices become free and no free white vertex is created.

Proof: Let $C = (c_i = 2, C_{-i})$ and $C' = (c'_i = 0, C_{-i})$. By Lemma 1, $u_i(C) = -4\lambda_1$. Note that all vertices in $\bar{N}_i$ are strongly dominated by v_i in C .

If at least two white vertices in $N_i^0(C)$ become free after v_i changes his strategy, then $u_i(C') \leq -4\lambda_2 < -4\lambda_1 = u_i(C)$, contradicting the fact that v_i is willing to change his strategy from 2 to 0.

If at least three gray vertices in $N_i^1(C)$ become free in C' after v_i changes his strategy, then $u_i(C') \leq -3\lambda_2 < -4\lambda_1 = u_i(C)$, a contradiction.

If exactly one white vertex in $N_i^0(C)$ and at least one gray vertex in $N_i^1(C)$ become free, then $u_i(C') \leq -3\lambda_2 < -4\lambda_1 = u_i(C)$, a contradiction. So, if a free white (resp. gray) vertex is created, then no gray (resp. white) vertex becomes free.

Combining the above arguments, the lemma is proved. ■

The following proofs are similar, using similar notations as in Lemma 12.

Lemma 13: In any strategy profile C , when a player v_i changes his strategy from 2 to 1, the number of strongly domi-

nated gray vertices is decreased by at most one and the number of white free vertices is not increased.

Proof: Because $BR(v_i, C) = 1$, one has $m_i(C') = 1$ by Lemma 2. By the definition of $m_i(C')$, one has $N_i^2(C') = \emptyset$, and thus $N_i^2(C) = \emptyset$. As a consequence, the black vertex v_i in C becomes a free gray vertex in C' . If the number of strongly dominated gray vertices is decreased by at least two, then at least two vertices in N_i that are gray in C become free in C' , and thus $u_i(C') \leq -\lambda_1 - 3\lambda_2 < -4\lambda_1 = u_i(C)$, a contradiction. If the number of white free vertices is increased by at least one, then at least one vertex in N_i that is white in C becomes free in C' , and thus $u_i(C') \leq -\lambda_1 - 3\lambda_2 < -4\lambda_1 = u_i(C)$, also a contradiction. These contradictions establish the lemma. ■

For simplicity of statement, by saying the *status* of a white vertex or a gray vertex, it refers to the situation of this vertex being strongly dominated or free.

Lemma 14: In any strategy profile C , when a player v_i changes his strategy from 1 to 0, the number of strongly dominated gray vertices is decreased by exactly one and the number of free white vertices is not increased.

Proof: Because $BR(v_i, C) = 0$, one has $m_i(C') = 0$ by Lemma 2, which implies that v_i has a neighbor of c -value 2 in C' , and thus in C . By Observation 1, $m_j(C) = m_j(C') \forall v_j \in V$, and thus the status of every $v_j \in V \setminus \{v_i\}$ is the same in both C and C' . Then, the lemma follows from the fact that the strongly dominated gray vertex v_i in C becomes a white strongly dominated vertex in C' . ■

Lemma 15: In any strategy profile C , when a player v_i changes his strategy from 0 to 1, the number of free white vertices is decreased by exactly one and the number of gray vertices that are strongly dominated is not decreased.

Proof: Because $BR(v_i, C) = 1$, one has $m_i(C') = 1$, and thus $N_i^2(C') = \emptyset$. Similar to the proof of the above lemma, the status of every $v_j \in V \setminus \{v_i\}$ is the same in both C and C' . Then, the lemma follows from the fact that the free white vertex v_i in C becomes a free gray vertex in C' . ■

Lemma 16: In any strategy profile C , when a player v_i changes his strategy from 0 to 2, one of the following three situations holds.

- 1) The number of white free vertices is decreased by at least two and the number of strongly dominated gray vertices is not decreased;
- 2) The number of strongly dominated gray vertices is increased by at least three and the number of free white vertices is not increased;
- 3) The number of free white vertices is decreased by exactly one and the number of strongly dominated gray vertices is increased by at least one.

Proof: If in C , there are at most two free gray vertices in $N_i^1(C)$ and no free white vertex in $\bar{N}_i^0(C)$, then $u_i(C) \geq -2\lambda_2 > -4\lambda_1 = u_i(C')$, a contradiction. So, there are at least three free gray vertices or at least one free white vertex that become strongly dominated in C' . In other words, one of the following two situations occurs: a) the number of strongly dominated gray vertices is increased by at least three, or b) the number of free white vertices is decreased by at least one.

Note that changing $c_i = 0$ to $c'_i = 2$ will never increase the

number of free white vertices. So, if a) occurs, then one has situation 2).

Next, consider the case when b) occurs. If the number of free white vertices is decreased by exactly one and the number of strongly dominated gray vertices is not increased, then there is exactly one free white vertex in $\bar{N}_i^0(C)$ and no free gray vertex in $N_i^1(C)$. It follows that $u_i(C) = -2\lambda_2 > -4\lambda_1 = u_i(C')$, a contradiction. So, if the number of free white vertices is decreased by exactly one, then one has situation 3).

If the number of free white vertices is decreased by at least two, then changing $c_i = 0$ to $c'_i = 2$ will never decrease the number of strongly dominated gray vertices, so one has situation 1). ■

Lemma 17: In any strategy profile C , when a player v_i changes his strategy from 1 to 2, one of the following three situations holds.

- 1) The number of free white vertices is decreased by at least one and the number of strongly dominated gray vertices is not decreased;
- 2) The number of strongly dominated gray vertices is increased by at least two and the number of free white vertices is not increased.
- 3) The number of free white vertices is decreased by at least two and the number of strongly dominated gray vertices is decreased by exactly one.

Proof: If there are at most two free gray vertices in $\bar{N}_i^1(C)$ and no free white vertex in $N_i^0(C)$, then $u_i(C) \geq -\lambda_1 - 2\lambda_2 > -4\lambda_1 = u_i(C')$, a contradiction. So one of the following two situations holds: a) there are at least three free gray vertices in $\bar{N}_i^1(C)$, or b) there are at least one free white vertex in $N_i^0(C)$.

If a) occurs, then the number of strongly dominated gray vertices is increased by at least two (note that if v_i is free, since it becomes black in C' , it cannot be counted as a new strongly dominated gray vertex). Combining this with the fact that changing $c_i = 1$ to $c_i = 2$ will never increase the number of free white vertices, one has situation 2).

Next, suppose b) occurs. Note that if v_i is a strongly dominated gray vertex in C and the number of free white vertices is decreased by exactly one, then $u_i(C) = -\lambda_1 - 2\lambda_2 > -4\lambda_1 = u_i(C')$, a contradiction. Also note that changing $c_i = 1$ to $c'_i = 2$ will not create new free gray vertex. Thus, the number of strongly dominated gray vertex can be decreased by at most one, and this decrease occurs only when v_i is a strongly dominated gray vertex in C . Combining these observations, if v_i is a strongly dominated gray vertex in C , then one has situation 3). If v_i is a free gray vertex in C , then one has situation 1). ■

Now, one is ready to prove Lemma 11.

Proof of Lemma 11: In the process of evolution from C to C' , where C and C' are the strategy profiles specified by the condition of Lemma 11, let x_{ij} be the number of players who have changed their strategies from i to j . Next, it will be proved that

$$x_{01} + x_{12} + 2x_{02} - x_{10} - x_{21} - 2x_{20} \leq 0. \quad (17)$$

In Lemma 12, there are two situations. Use x_{20}^1 and x_{20}^2 to represent the numbers of times that situation 1) and situation

2) occurred, respectively. Then, $x_{20} = s_{20}^1 + x_{20}^2$. Similarly, $x_{21} = x_{21}^1$, $x_{10} = x_{10}^1$, $x_{01} = x_{01}^1$. Note that the three situations in Lemma 16 and the three situations in Lemma 17 may have some overlap, so $x_{02} \leq x_{02}^1 + x_{02}^2 + x_{02}^3$ and $x_{12} \leq x_{12}^1 + x_{12}^2 + x_{12}^3$.

By Lemmas 12 to 17, the number of white free vertices created in the process is at most $x_{20}^1 - x_{01}^1 - 2x_{02}^1 - x_{02}^3 - x_{12}^1 - 2x_{12}^3$, and the number of strongly dominated gray vertices in the process is decreased by at most $2x_{20}^2 + x_{21}^1 + x_{10}^1 - 3x_{02}^2 - x_{02}^3 - 2x_{12}^2 + x_{12}^3$.

Because C' is an NE, by Corollary 1, there is no white free vertex in C' and any gray vertex in C' is free. Because C is an RDF, there is no white free vertex in C' . Let y be the number of strongly dominated gray vertices in C . Then,

$$x_{20}^1 - x_{01}^1 - 2x_{02}^1 - x_{02}^3 - x_{12}^1 - 2x_{12}^3 \geq 0 \quad (18)$$

$$2x_{20}^2 + x_{21}^1 + x_{10}^1 - 3x_{02}^2 - x_{02}^3 - 2x_{12}^2 + x_{12}^3 \geq y. \quad (19)$$

Adding (19) and (18), one has

$$x_{20}^1 + 2x_{20}^2 + x_{21}^1 + x_{10}^1 - x_{01}^1 - 2x_{02}^1 - 3x_{02}^2 - 2x_{02}^3 - x_{12}^1 - 2x_{12}^2 - x_{12}^3 \geq y. \quad (20)$$

Combining this with the relationship between x_{ij} and x_{ij}^k , one obtains

$$x_{01} + x_{12} + 2x_{02} - x_{10} - x_{21} - 2x_{20} \leq -y - x_{20}^1 - x_{02}^2 - x_{12}^2.$$

Since the variables y and x_{ij}^k 's are all non-negative, inequality (17) is proved, which completely verifies Lemma 11. ■

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